Aulia Kharis Rakhmasari

Zürich, Switzerland | arakhmasari@ethz.ch
github.com/auliakharis | linkedin.com/in/aulia-kharis-rakhmasari

Profile

MSc Data Science candidate at ETH Zürich working on LLM evaluation and alignment for regulated domains: LLM-as-a-Judge methodology, preference optimisation (DPO/LoRA), and benchmark design for financial and legal NLP. Previously 2 years as a production Data Scientist in banking, shipping ML pipelines with measurable revenue impact.

Technical Skills

ML & LLMs	PyTorch, Hugging Face (Transformers, PEFT, TRL), LoRA, DPO/RLHF, LLM-as-a-Judge evaluation, benchmark design, VLMs, TensorFlow
Infrastructure	SLURM / Euler HPC, Git, Bash, Cloudera Data Science Workbench
Languages	Python, R, SQL, C++
Data & BI	Advanced SQL at scale, Power BI, Tableau, R Shiny, Excel/VBA

Education

ETH Zürich **Zürich, Switzerland**
MSc Data Science, DINFK *Sep 2025 – expected Sep 2027*

- Teaching Assistant for Python and Machine Learning : D-ITET
- Fully funded by the Indonesian Government (LPDP) scholarship; acceptance rate $< 5\%$.
- Coursework: Deep Learning, Probabilistic AI, Machine Learning in Finance, Algorithmic Foundations of Data Science, Computational Statistics.

Institut Teknologi Sepuluh Nopember (ITS) **Surabaya, Indonesia**
BSc Statistics — GPA 3.97/4.00 *Jul 2019 – Jul 2023*

- **Best Graduate** among 1,000+ graduates; highest GPA in the 2019 cohort.
- Fully funded AIG Insurance Science Scholarship (semesters 3–8).
- Thesis: Marked Log-Gaussian Cox Process for estimating household expenditure per capita, Central Java.

Uniwersytet Szczeciński **Szczecin, Poland**
Exchange semester, Economics and Management *Sep 2021 – Feb 2022*

- Fully funded by the European Union under Erasmus+ International Credit Mobility.

Publications & Peer Review

- **V-FiLLM: Verified Financial LLM Reasoning Benchmark** arXiv:2608.11047
- **FinRegQA-EU: Corruption-Based Preference Data for Grounded EU Financial Regulatory Question Answering** arXiv:2609.36856
- **Investigating Relational Reasoning in VLMs** arXiv:2608.23518
- **Reviewer**, FinNLP @ EMNLP 2026.

Research Experience

FinRegQA-EU: Corruption-Based Preference Data for Grounded EU Financial Regulatory Question Answering
Law, Economics and Data Science Group (Prof. Elliott Ash), ETH Zürich *2026*

- Built an end-to-end evaluation and alignment pipeline on ETH's Euler HPC cluster over a purpose-built EBA/ESMA regulatory Q&A dataset.
- Ran multi-judge LLM-as-a-Judge evaluation across three judge models (Gemma-4-31B, Qwen3.5-27B, GLM-4.7-Flash); measured inter-judge agreement and found near-zero Krippendorff's α on citation quality despite strong agreement on accuracy.
- Fine-tuned Qwen3.5-4B with DPO across five preference-corruption strategies and introduced a **Citation F1** metric; showed that LLM judges are systematically blind to citation improvements that Citation F1 captures.

V-FiLLM — Verifiable Financial Reasoning Benchmark
ETH AI Center × Apertus (supervisor: Dr. Nino Antulov-Fantulin) *2026*

- Designed a deterministic computation-tree benchmark generating financial reasoning tasks with verifiable ground truth over structured tabular data.

- Benchmarked eight open and frontier models (incl. Apertus-70B, Llama-3.3-70B, Qwen3.5-27B) and ran LoRA fine-tuning experiments to quantify achievable gains.
- Stress-tested robustness under adversarial perturbations: missing values, OCR lookalike substitutions, prompt injection, and cross-table contamination.

Gated Graph-of-Gradients DPO — Fairness in LLM Alignment
ETH AI Center *2026*

- Investigated whether DPO loss disparities track demographic groups using the PRISM dataset.
- Showed that demographic grouping does *not* explain loss variation, whereas gradient-space grouping does; built a graph-of-gradients reweighting method on that finding.

Relational Reasoning in Vision-Language Models
ETH Zürich — completed *Sep 2025 – Jan 2026*

- Evaluated state-of-the-art VLMs on spatial, semantic, and compositional relational tasks; identified failure modes and proposed targeted evaluation benchmarks.

Research Assistant — Spatial Statistics
Institut Teknologi Sepuluh Nopember *Jun 2022 – Jun 2023*

- Marked LGCP modelling in R under Dr. A. Choiruddin, Dr. Sutikno, and Prof. J. Mateu (Universitat Jaume I)

Industry Experience

Data Scientist (full-time) **Jakarta, Indonesia**
Bank Syariah Indonesia *Sep 2023 – Sep 2025*

- Built and automated end-to-end ML pipelines in Python (Cloudera CDSW) for credit card programmes supporting over CHF 28M in annual card spend.
- Developed propensity models to reactivate dormant users and onboard new-to-bank customers, generating CHF 1.7 M+ in incremental transaction volume.
- Shipped an automated Next Best Offer recommendation engine for monthly personalised targeting.
- Forecast ATM and teller traffic through peak season at $< 5\%$ MAPE, integrated into live dashboards.

Data Scientist (part-time) **Jakarta, Indonesia (remote)**
Siena Clinical *May 2024 – Present*

- Built an interactive meta-analysis web application in R Shiny for real-time clinical data visualisation.
- Analysed clinical datasets for a study classifying benign prostatic hyperplasia vs. carcinoma; delivered internal training on meta-analysis in R.

Claim Data Analyst Intern **Jakarta, Indonesia**
AIG Insurance *May 2023 – Aug 2023*

- Automated claim reporting via Power BI dashboards and Excel VBA tooling; supported risk assessment on incoming claims.

Data Audit Analytics Intern **Jakarta, Indonesia**
PT XL Axiata Tbk *Feb 2022 – Jun 2022*

- Built KPI monitoring dashboards and presented audit findings on retail outlet and field survey data to senior management.

Selected Awards & Honours

- **1st place**, TrackAML Hackathon, PPATK (Indonesian Financial Transaction Reports and Analysis Centre). 2023
- **Distinction Graduate**, Bangkit Academy (Google, GoTo, Traveloka) — top 10% of 4,500+ participants, machine learning track. 2023
- **AWS Datathon Side Challenge**, ETH Zürich. 2026
- **Gold Medal**, ASEAN Innovative Science & Environmental Entrepreneur Fair; **Silver Medal**, World Science Environment & Engineering Competition. 2021–2022
- **Fully funded scholarships:** LPDP (Indonesian Government, ETH MSc); Erasmus+ International Credit Mobility (EU); AIG Insurance Science Scholarship.

Additional

- **Languages:** Indonesian (native), English (fluent, C1), German (A2).
- **Community:** Member, Data Science Indonesia (2021–present); President, Statistics Computer Course, ITS (2022).